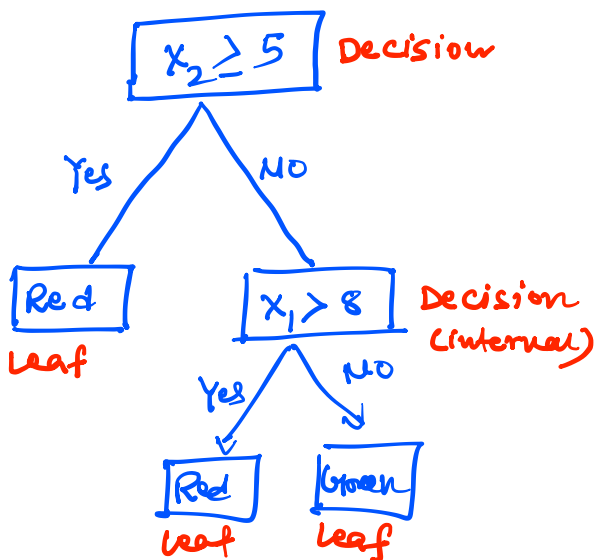
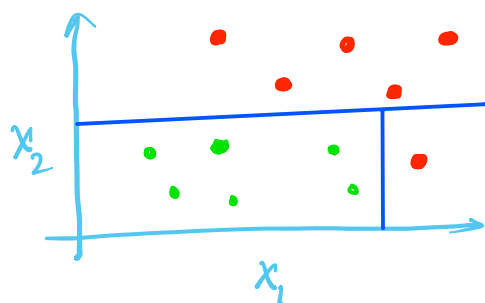


Decision Tree

Dataset
in samples.

Can solve both
classification &
Regression problems



categorical feature: question on feature == category?

numerical " : question on feature > threshold.

At leaf node:

for classification :- majority voting of class in node.

for regression :- Average of the samples in node.

Training:

- ① which feature gives max info.
- ② what rows goes into each child.
- ③ should each child be split again or declared as leaf.
- ④ At leaf node, how should we predict?

- ① 6 yes 6 no $\rightarrow$ 50-50 chance (Impure completely)
- ② 10 yes 2 no $\rightarrow$ yes class. (Purity somewhat)
- ③ 12 yes 0 no $\rightarrow$ yes (completely pure)

① Entropy as measure of uncertainty,

$$H(Y) = - \sum_{k=1}^K p_k \log_2(p_k)$$

$Y = \text{target}$

$k = \text{classes}$

$p_k = \frac{\text{\# samples in } k}{\text{\# samples}}$

$H(Y) = \text{entropy in } \underline{\text{bits}}$

9 yes 5 no

$$p_{\text{yes}} = \frac{9}{14}, \quad p_{\text{no}} = \frac{5}{14}$$

$$H(k) = - \frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

10 yes, 0 no

$$p_{\text{yes}} = \frac{10}{10} = 1, \quad p_{\text{no}} = \frac{0}{10} = 0$$

$$H(k) = - 1 \log_2 1 - 0 \log_2 0$$

$$= -1 \times 0 - 0 = 0$$

Gini Impurity:

$$\text{Gini}(Y) = 1 - \sum_{k=1}^K P_k^2$$

9 yes, 5 no.

$$= 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2$$

$$= 0.46$$

10 yes, 0 no.

$$= 0$$

* Information gain:

$$\text{IG}(Y; A) = H(Y) - H(Y|A)$$

Y = Target

A = feature (candidate)

$H(Y)$ = entropy of parent (before split)

$H(Y|A)$ = entropy after split on feature A .

$$H(Y|A) = \sum_{v \in \text{values}(A)} \frac{|D_v|}{|D|} H(D_v)$$

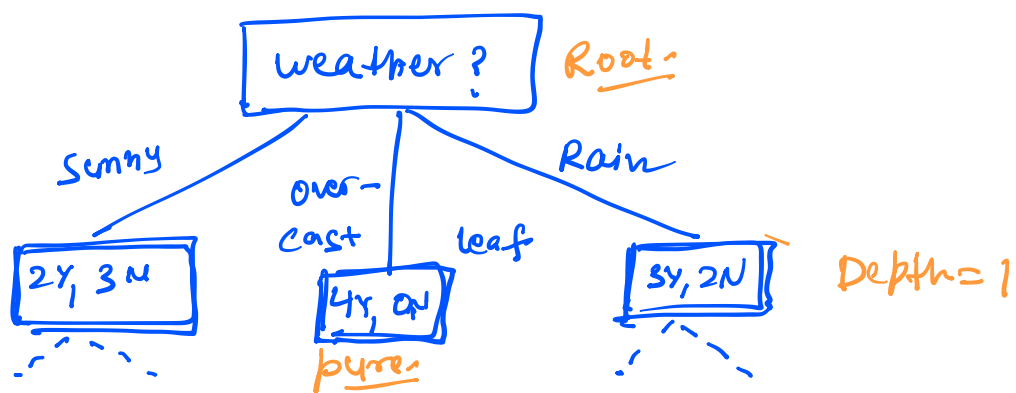
Day	Weather	Temperature	Humidity	Wind	Play cricket?
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Weather	Rows	Play Yes	Play No	Entropy	Weight
Sunny	5	2	3	0.971	5/14
Overcast	4	4	0	0.000	4/14
Rain	5	3	2	0.971	5/14

$$\begin{aligned}
 H(\text{play}/\text{weather}) &= \left(\frac{5}{14} \right) \times 0.971 + \left(\frac{4}{14} \right) (0) + \left(\frac{5}{14} \right) \times 0.971 \\
 &= 0.694
 \end{aligned}$$

↑ weight
↑ entropy

$$\begin{aligned}
 I_G(\text{play}/\text{weather}) &= 0.94 - 0.694 \\
 &= 0.246 \text{ bits}
 \end{aligned}$$



we take the feature to split on, which gives
 $\max^m \underline{IG}$.

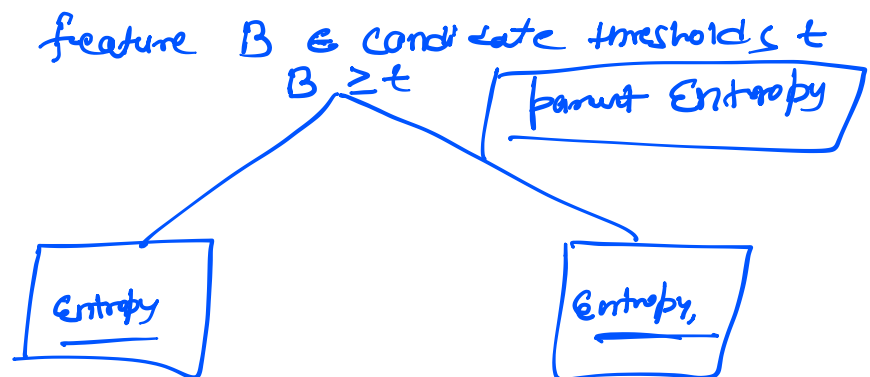
$$A^* = \arg \max_A IG(Y; A)$$

* threshold in numerical feature:

0, 2, 4, 6, 8, 10, 12.

$$\text{Candidate threshold} = \frac{0+2}{2} = 1,$$

1, 3, 5, 7, 9, 11



* when to stop recursion:

- ① no information gain at split
- ② pure node
- ③ no remaining features.
- ④ too few samples.
- ⑤ reached maximum depth.

* Inference :

new sample with all the features.

Trace the tree. Reach the leaf node.

Pure node $\rightarrow$ take the class.

Impure node $\rightarrow$ majority voting, Average

* Leaf probabilities :

Leaf node with 7 yes, 2 no.

$$P(\text{yes}) = \frac{7}{9},$$

$$P(\text{no}) = \frac{2}{9}.$$

* Depth decides model performance:

high depth $\Rightarrow$ overfitting.

low depth $\rightarrow$ underfitting,

* Feature Scaling :

Scaling is not necessary,

$$x_i < t$$

t is independent of other feature values.